FOCEX agent

User Manual

Force constants and phonons from a crystal structure: VASP · SC-SNAPS · FOCEX · Diagnosis & repair

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https://github.com/KeivanS/Anharmonic-lattice-dynamics (FOCEX/AGENT)

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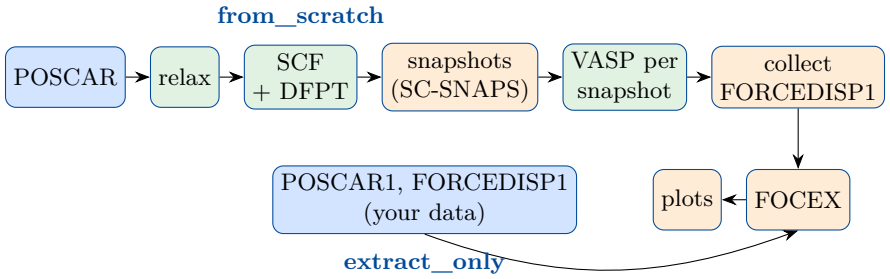
1 What the FOCEX agent does

FOCEX obtains the force constants of a crystal (harmonic, cubic, quartic, ...) by fitting them to the forces that density-functional theory (DFT) gives on the atoms of a supercell in which every atom has been displaced a little at random (a *snapshot*). From the force constants it computes the phonon dispersion, density of states, group velocities, Grüneisen parameters, elastic constants, sound speeds and the thermal expansion; THERMACOND then computes the lattice thermal conductivity from the same files.

Making those forces takes many steps: relax the crystal, orient the cell the way FOCEX expects, compute the Born charges of an insulator, choose a supercell, generate the snapshots, run one DFT calculation per snapshot, collect the forces, write the five input files of FOCEX and run it. Each step has its own input files and its own ways to fail. The FOCEX agent, `focex-agent.py`, does them all from one short *instruction file*, checks the result of every step, writes what it did to `report.log`, and can be restarted at any time: finished steps are not repeated. When a run stops, `diagnose.py` finds out where and why, and writes a script that repairs it (Section 10).

It works in one of two modes:

Mode	You give	The agent does
from_scratch	a crystal structure (`POSCAR`) and the instruction file	VASP relaxation, SCF and Born charges; SC-SNAPS snapshots; one VASP run per snapshot; FOCEX; plots (Section 4)
extract_only	existing supercells and their forces (`POSCAR1`, `FORCEDISP1`, ...) and the instruction file	rotates the data into the FOCEX frame; FOCEX; plots (Section 5)



green: VASP; orange: SC-SNAPS, FOCEX and their tools; blue: your input

2 Installation, step by step

Do the steps below in order, once on every computer. The same steps work on a Mac, on a Linux workstation and on the login node of a cluster; where they differ, the box says so.

Explanation: how to read this manual

Blue boxes hold what you type, one command per line. Everything after a `#` is a comment and need not be typed. **Amber boxes** like this one explain the step: what it does, and what to do if it fails. They are not commands.

2.1 What you need

What	Needed for	Step
gfortran, MPI, gnuplot, git	building the codes; running VASP in parallel; the plots	1
FOCEX, `read_outcar.x`, SC-SNAPS	the fit; reading the OUTCARs; the snapshots	2
VASP-Flow	writing the VASP inputs and batch scripts	3
Python $\geq$ 3.8 with numpy, spglib, pymatgen	the agent and `diagnose.py`	4
VASP 5.4 or newer and its POTCAR library	every DFT step (from_scratch only)	your licence; found in step 5

Step 1. The system tools

Do this — macOS (with Homebrew, https://brew.sh)

```
xcode-select --install                # git and the command-line tools
brew install gcc open-mpi gnuplot     # gfortran, mpif90 and mpirun, gnuplot
```

Do this — Linux (Debian, Ubuntu)

```
sudo apt install git gfortran libopenmpi-dev openmpi-bin gnuplot python3-venv
```

Do this — a cluster

```
module avail gcc openmpi python       # the names differ from cluster to cluster
module load gcc openmpi python        # put this line in ~/.bashrc
```

Explanation

gfortran and MPI are needed to build the codes (step 2), MPI also to run VASP on several cores. Without gnuplot the agent writes the plot scripts and you run `makeplots.sh` later, on a computer that has gnuplot.

Step 2. FOCEX, SC-SNAPS and the agent

Do this

```
cd ~
git clone https://github.com/KeivanS/Anharmonic-lattice-dynamics.git
cd Anharmonic-lattice-dynamics
make focex sc_snaps python-tools      # about 3 minutes
```

Explanation

The programs go to `~/ALADYN/bin` (the `DESTDIR` of `make.inc`). `python-tools` puts there the links `focex-agent` and `focex-diagnose` to the two Python scripts of `FOCEX/AGENT`. A plain `make` builds also THERMACOND (thermal conductivity) and ANFOMOD. To update later: `git pull`, then the same `make` line.

Step 3. VASP-Flow

Do this

```
cd ~
git clone https://github.com/KeivanS/VASP-Flow.git
```

Explanation

The agent uses VASP-Flow as a library to write the VASP input files and batch scripts; you do not need to learn VASP-Flow. It is looked for in `~/VASP-Flow`; if you put it elsewhere, step 5 records that place.

Step 4. The Python environment

Do this

```
python3 -m venv ~/myenv
~/myenv/bin/pip install numpy spglib pymatgen matplotlib
```

Explanation

One environment, `~/myenv`, serves both VASP-Flow and the agent. It is needed whenever *you* start `focex-agent.py` or `diagnose.py`: activate it first (`source ~/myenv/bin/activate`). The scripts the agent writes (`run_workflow.sh`, `resubmit_failed.sh`, `diagnosis/fix_it.sh`) contain the full path of the Python you used, so batch jobs need no activation.

Step 5. Let the configuration script find everything

Do this

```
source ~/myenv/bin/activate
cd ~/Anharmonic-lattice-dynamics/FOCEX/AGENT
bash focex-configure.sh               # 1. look and report only
bash focex-configure.sh --install     # 2. after "Ready": make it permanent
```

Explanation

`focex-configure.sh` finds out what this computer has and where: the operating system and processor, the number of cores, SLURM, gfortran and mpif90, the FOCEX, SC-SNAPS and `read_outcar.x` programs, Python and its packages, VASP-Flow, the POTCAR library (it searches your home folder and `/opt`, `/usr/local`, `/share`, `/apps`, `/sw` for folders named `potpaw_PBE*` or `PAW_PBE*`), `vasp_std`, MPI and gnuplot. Every item is reported as **ok**, **WARN** (optional, or needed only for some calculations) or **FAIL** (needed). For each problem the end of the report gives the command that fixes it.

It writes the settings, and only these, to `~/.focex-agent.env`: `VASP_POTCAR_DIR`, `VASP_STD`, `MPI_LAUNCH`, `VASP_FLOW_DIR` and `~/ALADYN/bin` added to `PATH` (Section 2.2). It then tests them: VASP-Flow builds a POTCAR from the library, and `focex-agent.py -version` starts. With `--install` it also adds one line to your shell start-up file (`~/.bash_profile` on a Mac with bash, `~/.bashrc` on Linux, `~/.zshrc` with zsh; a copy of the old file is kept) so that every new terminal and every batch job reads `~/.focex-agent.env`. Running it again is safe: the line is not added twice, and the previous settings are kept as `~/.focex-agent.env.bak`.

If it finds the wrong thing, or nothing, tell it: `--potcar /path/to/potpaw_PBE.54` (the folder with `Si/`, `O/`, ...), `--vasp /path/to/vasp_std`, `--python /path/to/python`.

Step 6. Check, in a new terminal

Do this

```
echo $VASP_POTCAR_DIR                  # the POTCAR library found in step 5
source ~/myenv/bin/activate
focex-agent --version                  # prints: focex-agent 1.4
```

Explanation

An empty first line means the start-up file was not read: on a Mac, check that the terminal runs the shell named by `echo $SHELL`, or type `source ~/.focex-agent.env`. `focex-agent: command not found` means `~/ALADYN/bin` is not in `PATH` (step 2 with `python-tools`, then step 5).

Step 7. Only on a cluster: a profile

Do this

```
cd ~/VASP-Flow/profiles
cp slurm.json mycluster.json
nano mycluster.json                    # VASP, modules, partition, time, account
```

Explanation

On a cluster VASP is started by SLURM, so the agent needs to know how: a *profile*, a small JSON file in `~/VASP-Flow/profiles/`. Set `vasp_std` (the VASP command, often just `vasp_std`), `modules` (what to load, e.g. `["gcc", "openmpi", "vasp/6.4.2"]`), `mpi_cmd` (`srun`), and in the `slurm` block `partition`, `nodes`, `ntasks_per_node`, `time` and `account` (your allocation; required). Then add `PROFILE: mycluster` to the instruction file, or give `--profile mycluster`. On a workstation no profile is needed: VASP is started with `MPI_LAUNCH` and `run_workflow.sh` with `bash`.

2.2 Settings outside the instruction file

These are environment variables; step 5 sets the first five for you.

variable	if not set	what it is
`VASP_POTCAR_DIR`	no POTCAR is made	the PBE POTCAR library: one folder per element
`VASP_POTCAR_DIR_LDA`	PBE folder with PBE→LDA in its name	the LDA library, for `FUNCTIONAL: LDA`
`VASP_STD`	`~/BIN/vasp_std`	the VASP executable (workstation; a profile overrides it)
`MPI_LAUNCH`	`mpirun -np`	the command before the number of ranks and VASP
`VASP_FLOW_DIR`	`~/VASP-Flow`	where VASP-Flow is
`VASP_OMP_NUM_THREADS`	1	OpenMP threads per MPI rank of VASP (box below)

Note. Threads. You do not have to do anything about OpenMP threads: the agent runs VASP with one thread per MPI rank (`OMP_NUM_THREADS=1`). Without that, a BLAS library built with OpenMP starts one thread per core in every MPI rank and the cores are oversubscribed many times (a 96-atom ZnO supercell made no SCF step in 30 minutes on 8 ranks). Only if your VASP is built for hybrid MPI + OpenMP runs and you want them, set `VASP_OMP_NUM_THREADS`.

3 Quick start

A from_scratch calculation of silicon on a workstation (Section 5 has extract_only).

1. A folder with the crystal structure

Do this

```
mkdir -p ~/work/Si && cd ~/work/Si
cp /path/to/your/POSCAR POSCAR
```

Explanation

The structure in VASP's POSCAR format, named `POSCAR` (another name: `--coord FILE`). Any cell of the crystal will do – primitive or not, any orientation: the agent reduces and orients it.

2. The instruction file

Do this

```
cat > focex-agent-instructions.txt <<'EOF'
Project: Si
RANKS: 1 1 1 0 0 0 0 0
MPI: 8
EOF
```

Explanation

This is the minimal file; Section 6 explains each line and every keyword that you may add. `MPI` is the number of cores VASP uses: `focex-configure.sh` prints a suggestion for this computer.

3. Set up the project

Do this

```
source ~/myenv/bin/activate
focex-agent                              # on a cluster: focex-agent --profile
    mycluster
```

Explanation

This takes seconds and runs no VASP. It creates the project folder (`Si`, after `Project:`; `PROJECT_DIR:` puts it elsewhere), with the relaxation inputs and `run_workflow.sh`, and prints the command that starts it. If the folder holds a project already, it stops; `--force` starts over. Before anything runs you can read, e.g., `Si/01_relax/INCAR`.

4. Start the workflow and follow it

Do this

```
cd Si
nohup bash run_workflow.sh > workflow.out 2>&1 &    # on a cluster: sbatch
    run_workflow.sh
tail -f report.log                                   # Ctrl-C stops the display,
    not the run
```

Explanation

`report.log` has one entry per task: OK or FAILED, the time, and what was checked (Section 4). Starting `run_workflow.sh` again continues where it stopped; finished tasks are not repeated.

5. If it stops

Do this

```
focex-diagnose ~/work/Si/Si
less ~/work/Si/Si/diagnosis/report.txt
bash ~/work/Si/Si/diagnosis/fix_it.sh
```

Explanation

`focex-diagnose` (`diagnose.py`) changes nothing itself; it writes what failed and why, and a script that repairs it (Section 10).

The options of `focex-agent` are all optional:

option	meaning
`--instr FILE`	the instruction file, if it is not called `focex-agent-instructions.txt`
`--coord FILE`	the crystal structure, if it is not called `POSCAR` (from_scratch)
`--profile NAME`	the VASP-Flow profile (step 7); default: `PROFILE:` of the instruction file, else none
`--force`	set up again a project folder that exists already. Its record of finished tasks is reset, so use it only to start over, after removing the old run folders

4 Mode from_scratch: from a POSCAR to phonons

`run_workflow.sh` runs `focex-agent.py run <PROJECT_DIR>`, which does the eleven tasks below in order. Each finished task is recorded in `state.json` and in `report.log`, with what was checked; a task that fails stops the workflow, and starting `run_workflow.sh` again continues from that task. A job that hits its time limit is simply submitted again.

The supercell tasks are checked again before they are skipped:

- **A removed or incomplete `04_supercells/snap_NNN` folder** is written again, and the `supercells` task and every task after it are redone. Only the folders without a converged OUTCAR run VASP.
- **To leave snapshots out on purpose**, move their folders to `04_supercells/unused/`. They are then not recreated, and `FORCEDISP1` is collected again from the others.
- **A `snap_NNN` folder with no `03_snaps/poscar_NNN`** stops the workflow.
- **To add snapshots**, raise SNAPSHOTS in the project's `focex-agent-instructions.txt` and start `run_workflow.sh` again. The extra snapshots are made at the same temperature and numbered after the existing ones, which are kept; only the new ones run VASP, then FOCEX is run again on all of them. A lower SNAPSHOTS removes nothing.

`relax` The structure you gave is reduced to its primitive cell (Niggli reduction) and rotated into the orientation FOCEX and SC-SNAPS use (Section 4.3); the result is `POSCAR.init`. VASP relaxes the atoms and the cell shape and volume in `01_relax`. If the forces are not

yet below EDIFFG after NSW ionic steps, the relaxation continues from its last structure (CONTCAR), up to three times.

poscar0 The relaxed cell is reduced and oriented again and written as `POSCAR0`: every later step uses this cell and this Cartesian frame. The report gives its space group and lattice parameters.

scf A self-consistent calculation on POSCAR0 in `02_scf` (same INCAR, no ionic steps, same k-mesh), which also gives the band gap. If the cell did not change in the previous task, it starts from the charge density and wave functions of the relaxation.

dfpt For an insulator (band gap above 0.05 eV, `DFPT: auto`): density-functional perturbation theory in `02_scf/dfpt` gives the high-frequency dielectric constant and the Born effective charges of every atom (Section 8). `DFPT: on` forces it, `off` skips it.

snaps The supercell is chosen (Section 4.2). SC-SNAPS (`03_snaps`) then writes the undistorted supercell `poscar_000` and 50 snapshots, each atom displaced as in a thermal ensemble of harmonic oscillators of frequency `AVG_FREQ` at temperature `TEMPERATURE`. **A temperature you give is used as it is**, e.g. to sample the displacements of that temperature for effective phonons; the report gives the resulting rms displacement. Only when `TEMPERATURE` is not given, the default 30 K is adjusted (SC-SNAPS rerun at most twice) until the rms displacement is within 15 % of `DISP_TARGET` (0.02 Å): large enough for clean forces, small enough for the Taylor expansion of the energy. The undistorted supercell is checked to be an exact multiple of POSCAR0 before any VASP job is run.

supercells One folder per snapshot, `04_supercells/snap_NNN`, with its POSCAR, the SCF INCAR without symmetry (`ISYM = 0`) and a k-mesh equivalent to the SCF one (Section 4.4); magnetic moments given per atom are carried over to the supercell.

vasp_snaps The VASP runs of the snapshots, one after the other. A run that fails does not stop the others; the failed ones are listed in `report.log`, and `resubmit_failed.sh` reruns only those. An existing OUTCAR is never overwritten (it is kept as `OUTCAR.1`, ...).

collect `read_outcar.x` reads the positions and forces of every run into `04_supercells/FORCEDISP1`; `POSCAR1` is the undistorted supercell. The report gives the largest net force on a supercell (should be ≈ 0) and the largest difference between the positions in OUTCAR and in POSCAR. The undistorted supercell (`snap_000`, run with `ISIF = 2`; the other snapshots use `ISIF = 1`, which gives only the trace) also gives the stress of the reference structure: `stress.params` (GPa, tensile positive, i.e. minus the `in kB` line of OUTCAR divided by 10) is written and copied to the FOCEX folders, where it enters the Huang rows and the elastic constants.

focex The five input files of FOCEX are written in `05_focex` (`structure.params`, `latdyn.params`, `kpbs.params`, `dielectric.params`, `default.params`) and FOCEX is run with the invariances held exactly (`enforce_inv = 1`). With Born charges a second run without them is made in `05_focex/NOBORN`, for comparison; with fewer than `LASSO_RATIO` (2) equations per unknown, a third run with LASSO and a refit (`enforce_inv = 3`), which keeps only the force constants the data need, is made in `05_focex/LASSO`. The report gives the fit error, the numbers of equations and unknowns, the lowest ω^2 (a warning below $-25\,\mathrm{cm}^{-2}$: imaginary modes) and the highest frequency. If the DOS range of `latdyn.params` does not fit the frequencies, FOCEX is run again with one that does.

`plots` `FOCEX/PLOTTING/allplots` fills the gnuplot scripts (bands, band Gruneisen parameters, group velocities, DOS, QHA, FCs) and makes the PDFs in every FOCEX folder (Section 9); `makeplots.sh` runs it again.

`thermacond` THERMACOND in `06_thermacond`: linewidths or the thermal conductivity, when `THERMACOND_JOB` is 1, 2 or 3; skipped for 0, the default (Section 11).

4.1 Defaults

- `ACCURACY: Accurate`: ENCUT 600 eV, EDIFF 10^{-7} eV, EDIFFG -0.001 eV/Å, PREC Accurate in every VASP step. `Normal`: 450 eV, 10^{-6}, -0.003, PREC Normal – cheaper, but the residual forces of the relaxed structure are larger (0.01 eV/Å in a ZnO test), which adds noise to the force constants. DFPT uses EDIFF 10^{-8}.
- Relaxation and SCF k-mesh: VASP-Flow's `coarse` density (1000 k-points per reciprocal atom), `fine` (6000) if every element is a metal. Each subdivision is rounded to the nearest integer; VASP-Flow itself rounds to even numbers (multiples of 6 in plane for hexagonal cells), the agent does not. The FOCEX mesh (`FOCEX_MESH`) is made the same way.
- 50 snapshots. Each gives three equations per atom of the supercell: 20 snapshots of a 96-atom supercell give 5760 equations for the 264 unknowns of ZnO with ranks 1–3, so fewer snapshots are often enough (`SNAPSHOTS`).
- `RANGE: 6 2 1 0 0 0 0`: neighbour shells kept for ranks 2, 3, 4, ...; `FC2RANGE: 0`: the harmonic force constants are kept over the whole supercell.
- FOCEX k-mesh `fine`; the band-structure path is found automatically.

4.2 The supercell

`SUPERCELL:` takes one of three forms:

- `SMALL` (the default): the multiple $m_1 \times m_2 \times m_3$ of the conventional cell with the fewest atoms whose faces are at least 10 Å apart, at most 100 atoms;
- `LARGE`: the same with 14 Å and at most 300 atoms. In both, if no such multiple is small enough, the one with the largest face distance; it keeps the symmetry of the lattice;
- **the supercell matrix itself**: nine integers, the three rows of the matrix that the agent writes to SC-SNAPS's `supercell.inp`. The first three integers $n_{11}\, n_{12}\, n_{13}$ give the first supercell vector as a combination of the primitive vectors of POSCAR0, $\mathbf{R}_{S1} = n_{11}\mathbf{R}_{01} + n_{12}\mathbf{R}_{02} + n_{13}\mathbf{R}_{03}$; the next three give $\mathbf{R}_{S2}$ and the last three $\mathbf{R}_{S3}$ in the same way (the agent's `cell.inp` has `convcoord = 1`, primitive units). For example `SUPERCELL: 2 0 0 0 2 0 0 0 2` is $2 \times 2 \times 2$ primitive cells, and for an fcc crystal `SUPERCELL: -1 1 1 1 -1 1 1 1 -1` is its conventional cube (4 primitive cells). The determinant must be positive.

The report gives the matrix and the number of atoms (for SMALL and LARGE also the face distance).

4.3 The FOCEX frame

FOCEX and SC-SNAPS build the conventional cell from $a, b, c, \alpha, \beta, \gamma$ with a along x and b in the xy plane, and the supercells, forces and Born charges must be in that same Cartesian frame. The agent rotates the cell into it at the start, so everything VASP computes is already in the right frame. `structure.params` lists the conventional cell (line 1), the primitive vectors in its units (line 2) and the atoms in conventional coordinates.

4.4 The k-mesh of the supercells

If the supercell is a diagonal multiple of the primitive cell, its mesh is the SCF mesh divided by the multiples, rounded up (6 6 4 with a $3\times3\times2$ supercell gives 2 2 2). Otherwise (every multiple of a centred conventional cell) it keeps the k-point spacing of the SCF mesh. `KPOINTS_snaps` sets it yourself (Section 7).

5 Mode extract_only: FOCEX on existing data

Use it when the supercell forces exist already – from an earlier calculation, another code, or a colleague – and you want FOCEX's force constants and phonons. No VASP and no SC-SNAPS are run.

The data folder (named in the instruction file with `DATA:`) holds

- `POSCAR1` and `FORCEDISP1`: a supercell and the positions and forces of its snapshots, in FOCEX's format (what `read_outcar.x` or `read_qe.x` write); more pairs `POSCAR2`/`FORCEDISP2`, ... are all used. Any orientation.
- optionally `POSCAR`: the primitive cell, in the same orientation as the data; without it, spglib finds the primitive cell of `POSCAR1`.
- optionally `dielectric.params`: the dielectric constant and the Born charges, in the frame of the data, one 3×3 block per atom of that primitive cell. With more than one atom of a species the primitive `POSCAR` is required, to fix the order of the atoms.

`SPECIES: Mg O` names the elements when the POSCARs have no element line.

What the agent does. At set-up it symmetrizes the lattice of the primitive cell (DFT lattice parameters given to 6–7 digits make FOCEX's symmetry analysis lose force constants), finds the rotation into the FOCEX frame, and rotates the supercells, the positions and forces in `FORCEDISPi`, the dielectric constant and the Born charges with it. The report gives the rotation and the size of the symmetrization. The workflow then has only the tasks focex and plots.

```
cat > focex-agent-instructions.txt <<'EOF'
Project: MgO
MODE: extract_only
DATA: ~/data/MgO_forces          # POSCAR1, FORCEDISP1, dielectric.params
SPECIES: Mg O
RANKS: 1 1 1 0 0 0 0 0
EOF
python ~/Anharmonic-lattice-dynamics/FOCEX/AGENT/focex-agent.py
cd MgO && bash run_workflow.sh          # FOCEX takes minutes: no batch job needed
```

Checked on MgO data given in a rhombohedral orientation: the fit error (7.197 %), the 35 unknowns, the frequencies and the elastic moduli are those of a FOCEX run in the original frame.

6 The instruction file

Explanation: the format

One keyword per line, `KEYWORD: value`; upper or lower case; `#` starts a comment. Blocks that change whole generated files are described in Section 7. A commented example of every line is `FOCEX/AGENT/focex-agent-instructions.txt`.

6.1 The minimal file

Do this — from_scratch: start from a POSCAR

```
Project: Si
RANKS: 1 1 1 0 0 0 0 0
MPI: 8
```

Explanation

`RANKS` **the only keyword that must be given**; the agent stops without it. Eight flags, one per rank of force constants 1–8: `1 1 0 0 0 0 0 0` harmonic, `1 1 1 0 0 0 0 0` harmonic and cubic (needed for Grüneisen parameters, thermal expansion, thermal conductivity), `1 1 1 1 0 0 0 0` up to quartic.

`Project` a name, used in titles and as the name of the project folder. Without it the name is `VASP_Calculation`.

`MPI` the number of cores (MPI ranks) VASP uses. **Without it VASP runs on one core**, also with a profile.

On a cluster add `PROFILE: mycluster` (step 7 of Section 2), and `ACCOUNT: <allocation>` if the profile has no account.

Do this — extract_only: FOCEX on forces you already have

```
Project: MgO
DATA: ~/data/MgO_forces
RANKS: 1 1 1 0 0 0 0 0
```

Explanation

`DATA` is the folder with `POSCAR1` and `FORCEDISP1` (Section 5); giving it selects `MODE: extract_only`. No VASP is run, so `MPI` is not needed.

6.2 Everything else: the defaults, and how to change them

Every keyword below may be left out; the second column is what the agent then does. To change it, add the line of the third column (with your value) to the instruction file.

keyword	if left out	to change it, e.g.
project		
`PROJECT_DIR`	folder named after Project, next to the instruction file	`PROJECT_DIR: /scratch/me/Si`
`MODE`	from_scratch; extract_only when DATA is given	`MODE: extract_only`
`SPECIES`	element names from the POSCARs (extract_only)	`SPECIES: Mg O`
`PROFILE`	none: a workstation, `bash run_workflow.sh`	`PROFILE: mycluster`
`ACCOUNT`	the account of the profile (SLURM)	`ACCOUNT: my_allocation`
force constants		
`RANGE`	6 2 1 0 0 0: neighbour shells for ranks 2–8	`RANGE: 8 3 2 0 0 0`

keyword	if left out	to change it, e.g.
FC2RANGE	0: harmonic range = the supercell	FC2RANGE: 0
LASSO_RATIO	2: below 2 equations per unknown an extra LASSO run	LASSO_RATIO: 3
VASP		
FUNCTIONAL	R2SCAN	FUNCTIONAL: PBEsol (PBE, PBEsol, LDA, AM05, R2SCAN, HSE06, VV10)
ACCURACY	Accurate: ENCUT 600 eV, EDIFF 10^{-7}, EDIFFG -0.001, PREC Accurate	ACCURACY: Normal (450 eV, 10^{-6}, -0.003)
ENCUT	from ACCURACY	ENCUT: 520
KMESH_DENSITY	coarse (1000 k-points per reciprocal atom); fine (6000) if every element is a metal	KMESH_DENSITY: fine (or a number)
MAGMOM	no magnetic moments	MAGMOM: Cr=3.0
GGA_U	auto: tabulated U on d/f elements if the compound has O, S, Se, Te or a halogen	GGA_U: OFF (or ON)
DFPT	auto: Born charges and ε_∞ if the SCF shows a gap (>0.05 eV)	DFPT: off (or on)
BORN_TRANSPOSE	yes (Section 8)	BORN_TRANSPOSE: no
snapshots		
SUPERCELL	SMALL: faces ≥10 Å apart, ≤100 atoms (Section 4.2)	SUPERCELL: LARGE or SUPERCELL: 2 0 0 0 2 0 0 0 2
SNAPSHOTS	50	SNAPSHOTS: 20
TEMPERATURE	30 K, adjusted until the rms displacement is DISP_TARGET	TEMPERATURE: 300 (used as given)
DISP_TARGET	0.02 Å (only without TEMPERATURE)	DISP_TARGET: 0.03
AVG_FREQ	400 cm^{-1}, frequency of the SC-SNAPS oscillators	AVG_FREQ: 250
FOCEX		
FOCEX_MESH	fine	FOCEX_MESH: 12 12 12 (or coarse, or a density)
KPATH	0: band path found automatically	KPATH: 1 (and a kpbs_params block)
BS_NKPTS	40 points per segment of the band path	BS_NKPTS: 60
THERMACOND (Section 11)		
THERMACOND_JOB	0: not run	THERMACOND_JOB: 2 (1 linewidths, 2 or 3 κ)
THERMACOND_MESH	the FOCEX mesh (usually far too dense)	THERMACOND_MESH: 12 12 12
THERMACOND_T	0 900 31: Tmin, Tmax (K), number of temperatures	THERMACOND_T: 100 900 9
THERMACOND_SOLVER	0: conjugate gradient (job 2)	THERMACOND_SOLVER: 1 (SVD)

keyword	if left out	to change it, e.g.
programs		
FOCEX_X	the newest ~/ALADYN/bin/focex-*.x	FOCEX_X: /ALADYN/bin/focex-9.8.x
SC_SNAPS_X	~/ALADYN/bin/sc_snaps.x (else ~/BIN/sc_snaps.x)	SC_SNAPS_X: /path/sc_snaps.x
READ_OUTCAR_X	~/ALADYN/bin/read_outcar.x	READ_OUTCAR_X: /path/read_outcar.x
THERMACOND_X	the newest ~/ALADYN/bin/tcond-*.x (MPI build)	THERMACOND_X: /path/tcond-2.0.x

Further VASP keywords are those of VASP-Flow (its manual lists them all); single INCAR tags are best given in an INCAR block (Section 7).

7 Changing the files the agent writes

Every file the agent generates can be changed *from the instruction file*, with a block named after the file: a line with its name, the contents, and a line with the name followed by _END.

- **INCAR blocks are merged tag by tag**: the tags you give replace or add to those the agent writes; the others stay. INCAR applies to every VASP step; INCAR_relax, INCAR_scf, INCAR_dfpt, INCAR_snaps to one.
- **Every other block replaces the whole file**, so it must be complete: KPOINTS (relaxation and SCF), KPOINTS_snaps, cell_inp, supercell_inp, snaps_inp (SC-SNAPS), structure_params, latdyn_params, kpbs_params, dielectric_params, default_params (FOCEX).

Simple choices – the supercell, the number of snapshots, the FOCEX k-mesh – have their own keyword; use a block only for what has none.

Example. Wurtzite ZnO without Hubbard U (VASP-Flow would otherwise put its tabulated U on Zn, an oxide), LASPH in every VASP step, stricter electronic convergence only in the supercell runs, a $3 \times 3 \times 2$ supercell, 20 snapshots, a denser FOCEX mesh, and your own band path Γ–M–K–Γ–A, which replaces kpbs.params:

```
Project: ZnO
RANKS: 1 1 1 0 0 0 0 0
FUNCTIONAL: PBEsol
GGA_U: OFF
MPI: 40
SUPERCELL: 3 0 0  0 3 0  0 0 2       # rows: supercell vectors in primitive units
SNAPSHOTS: 20
FOCEX_MESH: 12 12 8
KPATH: 1                              # the band path is given below

INCAR                                 # every VASP step (merged tag by tag)
  LASPH = .TRUE.
INCAR_END
INCAR_snaps                           # the supercell runs only
  NELM = 200
  ALGO = All
INCAR_snaps_END
```

```
kpbs_params                          # replaces kpbs.params (FOCEX's band path)
1        # coordinates in units of the primitive reciprocal vectors
60       # k-points per segment
4        # number of segments
G   0.000000 0.000000 0.000000
M   0.500000 0.000000 0.000000
K   0.333333 0.333333 0.000000
G   0.000000 0.000000 0.000000
A   0.000000 0.000000 0.500000
kpbs_params_END
```

After the set-up you can read `01_relax/INCAR` before anything runs.

8 Born charges and the dielectric constant

In an insulator, the long-range Coulomb interaction between the displaced ions splits the longitudinal and transverse optical modes near Γ (LO–TO splitting). FOCEX includes it with the Born effective charges Z^* and the high-frequency dielectric constant ε_∞; without them the optical branches near Γ are wrong.

- from_scratch: the agent computes them (task dfpt) when the SCF shows a band gap, writes them to `dielectric.params`, and makes for comparison a FOCEX run without them in `05_focex/NOBORN`. `DFPT: off` skips them; `on` computes them for any material.
- extract_only: give them in `dielectric.params` in the data folder.
- your own values, in either mode: a `dielectric_params` block (it replaces the file).

Note. VASP prints Z^*_{ij} with i the direction of the field and j that of the displacement; FOCEX expects the transpose, which the agent writes. `BORN_TRANSPOSE: no` writes them as VASP prints them; you should not need it.

9 The project folder and the results

Figure 1 shows the folders the agent makes; the table lists what each holds.

file or folder	what it is
`focex-agent-instructions.txt`	the copy of your instruction file that the runs read
`report.log`	one entry per task: OK or FAILED, minutes, what was checked – read this first
`state.json`	the finished tasks and the numbers passed between them
`run_workflow.sh`	the job; `resubmit_failed.sh` reruns the failed supercell runs
`POSCAR.input, POSCAR.init, POSCAR0`	your structure; reduced and oriented; relaxed
`01_relax, 02_scf, 02_scf/dfpt`	VASP: relaxation, SCF, Born charges
`03_snaps`	SC-SNAPS input and the snapshots `poscar_NNN`
`04_supercells`	`snap_NNN/` (one VASP run each), `POSCAR1`, `FORCEDISP1`
`05_focex`	FOCEX: inputs, outputs, plots; `NOBORN/` and `LASSO/` when made

file or folder	what it is
`06_thermacond`	THERMACOND (`THERMACOND_JOB` 1–3): `kappa_vs_T.dat`, `kappa_wigner.dat`, ...
`diagnosis`	written by `diagnose.py`: `report.txt`, `fix_it.sh`, backups

In `05_focex`: the force constants `fc2.dat`, `fc3.dat`, ... with the cell in `lat_fc.dat`; the dispersion `bs_freq.dat`; the elastic constants and moduli `mech.dat`; thermodynamics and thermal expansion `thermo_QHA.dat`; and the PDFs (`<Name>` is the project folder): `bands-<Name>.pdf`, `freq_bands-<Name>.pdf` (one colour per branch), `bandgrun-<Name>.pdf` (dispersion above the Grüneisen parameters along the path), `bvel-<Name>.pdf` (the dispersion coloured by group velocity, and the group velocities against frequency), `all-<Name>.pdf` (Grüneisen parameters, density of states, dispersion coloured by the Grüneisen parameters), `qha.pdf` and `thermo.pdf` (thermal expansion, Grüneisen parameter, bulk modulus, thermodynamic functions, free energy against volume), `fcs-<Name>.pdf` and `chi.pdf`. The FOCEX manual describes every output file. The large VASP files (WAVECAR, CHGCAR, CHG, ...) are deleted when no longer needed; the OUTCARs never.

10 When a run stops: diagnose.py

```
A=~/Anharmonic-lattice-dynamics/FOCEX/AGENT
python $A/diagnose.py ~/work/ZnO/ZnO      # one or more project folders
less ~/work/ZnO/ZnO/diagnosis/report.txt
bash ~/work/ZnO/ZnO/diagnosis/fix_it.sh
```

`diagnose.py` itself changes nothing. It lists every task as done, FAILED, INTERRUPTED (the job was killed inside it) or not reached, or reports that the project is still running or not yet started. For a failed task it finds the cause from `report.log`, the OUTCARs, `vasp.out`, `focex.out` and the SLURM logs, and writes `diagnosis/report.txt` (what failed, where, the evidence, what the fix will change) and `diagnosis/fix_it.sh`. VASP failures are recognised with the rules of VASP-Flow's diagnosis tool; FOCEX, SC-SNAPS and `read_outcar.x` stops by their messages.

cause	meaning	what fix_it.sh does
`timeout`	the job hit its time limit	walltime ×2, resubmit
`scf_nelm`	SCF not converged in NELM steps	ALGO = All, NELM = 200
`edddav`	the Davidson eigensolver failed	ALGO = All
`zbrent`	the relaxation's line search failed	IBRION = 1
`relax_unconv`	relaxation not converged	resubmit (continues from CONTCAR)
`oom`	out of memory	KPAR = 1
`mpi_init`, `not_run`, `interrupted`	MPI did not start; never ran; killed	resubmit
`others`	disk quota, symmetry, an unknown stop, ...	held, with the error lines

New INCAR tags go into the INCARs of the failed runs and into the matching block of the project's instruction file (`INCAR_snaps`, ...), so that later runs use them too. Before it changes anything, `fix_it.sh` checks that nothing is running and that nothing changed since the diagnosis, and copies every file it edits to `diagnosis/backup_<date>/`; then it resubmits (`sbatch` with a SLURM profile, otherwise `bash` in the background). When the fix needs your

The project folder (`focex-agent.py`, mode `from_scratch`)

folder written at set-up written by the runs made by you written by the repairs

`~/calc/`	where you call `focex-agent.py`
├─ `focex-agent-instructions.txt, POSCAR`	your input
└─ `Si/`	project folder (`PROJECT_DIR`; default `PROJECT_NAME`)
├─ `focex-agent-instructions.txt`	the copy the runs read
├─ `POSCAR.input, POSCAR.init, POTCAR`	your structure; reduced and oriented
├─ `run_workflow.sh, state.json`	the job (`sbatch` or `bash`); its state
├─ `resubmit_failed.sh`	reruns failed supercell runs
├─ `report.log, POSCAR0`	one entry per task; the relaxed cell
├─ `01_relax/`	relaxation
│ ├─ `INCAR KPOINTS POSCAR POTCAR run.sh`	
│ └─ `OUTCAR OSZICAR CONTCAR vasprun.xml`	
├─ `02_scf/`	SCF on POSCAR0
│ ├─ `INCAR KPOINTS POSCAR POTCAR; OUTCAR ...`	
│ └─ `dfpt/`	Born charges, ϵ^{∞} (only with a gap)
│ └─ `run.sh copy_from_scf.sh extract_born.py; OUTCAR`	
├─ `03_snaps/`	SC-SNAPS: the supercell and the snapshots
│ ├─ `cell.inp supercell.inp snaps.inp`	
│ ├─ `poscar_000 ... poscar_NNN`	snapshots; `poscar_000` undisplaced
│ ├─ `snapshots.xyz`	all snapshots, for viewing
│ └─ `added_NNN-MMM/`	when `SNAPSHOTS` is raised
├─ `04_supercells/`	the supercell VASP runs
│ ├─ `snap_000/ ... snap_NNN/`	INCAR KPOINTS POSCAR POTCAR; OUTCAR
│ ├─ `unused/`	snapshots you set aside
│ └─ `FORCEDISP1, POSCAR1, stress.params`	forces; supercell; stress of `snap_000`
├─ `05_focex/`	FOCEX
│ ├─ `*.params, POSCAR1, FORCEDISP1`	the five inputs and the data
│ ├─ `fc*.dat lat_fc.dat log*.dat`	force constants and the log
│ ├─ `bs_freq.dat mech.dat thermo_QHA.dat ...`	phonons, elastic constants, QHA
│ ├─ `makeplots.sh, *-Si.pdf`	the plots
│ ├─ `NOBORN/`	with Born charges: the fit without them
│ └─ `LASSO/`	few equations per unknown (`LASSO_RATIO`)
├─ `06_thermacond/`	THERMACOND (`THERMACOND_JOB` 1–3)
│ └─ `lat_fc.dat fc2.dat fc3.dat *.params`	`kappa_vs_T.dat`, ...
└─ `diagnosis/`	`diagnose.py`: report.txt, fix_it.sh

Figure 1: The folders made from where you call `focex-agent.py` (here `~/calc`, with `PROJECT_NAME: Si`). `focex-agent.py` writes the project folder, `01_relax/`, `run_workflow.sh` and `resubmit_failed.sh` at once; every other step folder is made when its task runs, in the order of the numbers. Each VASP folder is an ordinary VASP folder. In mode `extract_only` there are no `01`–`03` folders: `04_supercells/` holds your data, rotated into the FOCEX frame, and `05_focex/` the fit.

judgement – a cause it does not know, a fix that was already tried, a new supercell – the script is *held*: it prints what to do and stops, and `bash fix_it.sh -anyway` runs its suggested commands as they are. Run `diagnose.py` again when the resubmitted job has finished.

11 Thermal conductivity

The last task, `thermacond`, runs THERMACOND on the force constants of `05_focex`. Whether it runs is decided by the THERMACOND job, line 8 of `latdyn.params`, which you set in the instruction file:

THERMACOND_JOB	what THERMACOND does
0 (default)	nothing: the task is skipped (FOCEX has already given the bands, DOS and Grüneisen parameters)
1	three-phonon linewidths of every mode along the band path (`lw-selfenergy.dat`)
2	thermal conductivity: RTA and the direct solution of the BTE
3	thermal conductivity: RTA and the iterative solution of the BTE

```
RANKS: 1 1 1 0 0 0 0 0        # cubic force constants are needed
THERMACOND_JOB: 2
THERMACOND_MESH: 12 12 12     # latdyn.params line 1 for THERMACOND (default: the
    FOCEX mesh)
THERMACOND_T: 100 900 9       # line 6: Tmin, Tmax (K), number of temperatures
# THERMACOND_SOLVER: 0        # line 9, job 2: 0 conjugate gradient, 1 SVD
```

The task copies `lat_fc.dat`, `fc2.dat`, `fc3.dat`, `structure.params`, `dielectric.params` and `kpbs.params` from `05_focex` to `06_thermacond`, writes there FOCEX's `latdyn.params` with lines 1, 6, 8 and 9 from these keywords, and runs `tcond-2.0.x` with the MPI command of the profile; `05_focex` is not changed. A `latdyn_params` block (Section 7) sets all 13 lines for both FOCEX and THERMACOND, and its line 8 is the job unless `THERMACOND_JOB` is given. The mesh is the expensive choice: the FOCEX default (`fine`) is usually far too dense, so set `THERMACOND_MESH` and converge `kappa` with it.

The report gives, at up to seven of the temperatures, κ_{RTA}, the full (direct or iterative) solution and the coherences term κ_C of the Wigner formulation (W/mK, average of the diagonal), and a warning if the full solution falls below the RTA, which it should not do (the direct solver has an open issue: compare with job 3). The complete results are in `06_thermacond`: `kappa_vs_T.dat`, `kappa_wigner.dat`, `Qvalue.dat` (scattering rates) and the other files described in the THERMACOND manual. An existing project gets the task by adding `THERMACOND_JOB` to its `focex-agent-instructions.txt` and submitting `run_workflow.sh` again: the finished tasks are skipped. Changing `THERMACOND_JOB`, `_MESH`, `_T` or `_SOLVER` later and submitting again runs the task again; V3·δ is read back from `06_thermacond/V3DELTA` when the force constants and the mesh are the same (job 3 after job 2, new temperatures).

12 Version history

1.4 2026-11-07: the FOCEX fits hold the invariances exactly (`enforce_inv = 1`; was 0) and the extra LASSO run uses the refit (`enforce_inv = 3`; was 2), as the FOCEX FAQ recommends; `diagnose.py` reports the same version.

1.3 2026-11-06: `focex-configure.sh` finds the programs, the POTCAR library, VASP and MPI and writes `~/.focex-agent.env`; installation and quick start as numbered steps (commands in blue boxes, explanations in amber boxes); the minimal instruction file and

a table of every default and how to change it; the manual now says that without `MPI` VASP runs on one core; a diagram of the project folder (Figure 1).

`1.2` 2026-10-27, with FOCEX 9.0: the undistorted supercell `snap_000` is run with `ISIF = 2`, and `collect` writes its stress, minus the `in kB` line of OUTCAR divided by 10, as `stress.params` (GPa, tensile positive) for FOCEX (Huang rows and elastic constants); the test VASP stand-in prints the stress.

`1.1` 2026-10-23: task `thermacond`, switched on by `THERMACOND_JOB` (latdyn.params line 8), with `THERMACOND_MESH`, `THERMACOND_T`, `THERMACOND_SOLVER`; a given `TEMPERATURE` is kept; `SUPERCELL` matrix and `PROJECT_DIR` documented; SC-SNAPS periodic images fixed.

`1.0` 2026-10-23, with FOCEX 8.19: modes from_scratch and extract_only; instruction file `focex-agent-instructions.txt`, `--instr`, `--coord`, `PROJECT_DIR`; `diagnose.py`; this manual.